$\mathcal{W}$-algebras and LCFT

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SUNY-Albany

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Today's talk

More precisely

- C_2-cofinite irrational vertex algebras
- Construction of logarithmic modules

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Conformal Vertex Algebras

All $\mathcal{W}$-algebras in this talk are conformal vertex algebras , that is vertex algebras equipped with an action of the Virasoro algebra.

Vertex algebras and vertex operators have applications in many areas of mathematics:

- Representation theory of affine Kac-Moody Lie algebras and related algebras
- Monstrous Moonshine (Frenkel-Lepowsky-Meurman, Borchards)
- Category theory (Huang-Lepowsky's tensor product theory)
- Geometric Langlands correspondence (E. Frenkel, Beilinson-Drinfeld)

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Rational and irrational vertex algebras

- A large part of the theory deals with strongly rational or regular vertex algebras (= the category of *all* modules is semisimple). Examples include certain affine vertex algebras, lattice vertex algebras, Virasoro minimal models, Moonshine module.
- Recent developments in physics, particularly in logarithmic conformal field theory, show that irrational vertex algebras with finitely many irreps are also of considerable interest.

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C_2-cofiniteness

Let V be a (conformal) vertex algebra. We will use

$$Y(u,x) = \sum_{n\in\mathbb{Z}} u_n x^{-n-1}$$

to denote the vertex operator associated to $u \in V$.

Vertex algebra V is said to be C_2-cofinite if the vector space

$$C_2(V) = \langle a_{-2}b : a, b \in V \rangle$$

is of finite codimension.

Theorem (Zhu)

If V is C_2-cofinite then it has finitely many irreps (up to equivalence).

Strongly rational vertex algebras are always C_2-cofinite.

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C_2-cofinite irrational vertex algebras

- The category of f.g. $V - mod$ can be endowed with a natural braided finite tensor category structure (Huang-Lepowsky-Zhang) , although not necessarily rigid.
- It is expected that $V - Mod \cong A - Mod$, where A is a finite-dimensional quantum group.
- The space of generalized characters of V-modules is modular invariant (Miyamoto).
- These algebras have rich combinatorial structure (still poorly understood).

All known examples of C_2-cofinite, irrational vertex algebras come from non-generic $\mathcal{W}$-algebras. No known examples coming from affine Lie algebras!

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Quantum $\mathcal{W}$-algebras

Different definitions in the literature. In this talk

Definition

(Quantum) $\mathcal{W}$-algebra is a conformal vertex algebra strongly generated by a set $\{w^1, ..., w^k\}$ of *primary vectors* (= annihilated by the Virasoro generators $L(n)$, $n \geq 1$) and ω, with non-linearities in the commutation relations.

If $\deg(w^i) = r_i$, then the $\mathcal{W}$-algebra is said to be of type $(2, r_1, ..., r_k)$.

Strongly generated: spanned by

$$w^{i_1}_{n_1} \cdots w^{i_k}_{n_k} \mathbf{1}, \quad n_i \leq -1 \quad .$$

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Quantum Drinfeld-Sokolov reduction

$\bar{\mathfrak{g}}$ - a finite-dimensional complex simple Lie algebra,

$\bar{\mathfrak{g}} = \bar{\mathfrak{n}}_+ + \bar{\mathfrak{h}} + \bar{\mathfrak{n}}_-$.

$\mathfrak{n}_\pm = \bar{\mathfrak{n}}_\pm \otimes \mathbf{C}[t, t^{-1}]$

$\mathfrak{g}$ - the Kac-Moody affinization of $\bar{\mathfrak{g}}$ and $k \neq -h^\vee$ the level of $\mathfrak{g}$.

$V_k(\bar{\mathfrak{g}}) = U(\mathfrak{g}) \otimes_{C[t]\otimes\bar{\mathfrak{g}}+\mathbb{C}c} \mathbb{C}$ - the universal (conformal) vertex algebra of level k.

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Quantum Drinfeld-Sokolov reduction (cont.)

Consider the zeroth cohomology group of the Drinfeld-Sokolov complex

$$\mathcal{W}^k(\bar{\mathfrak{g}}) := H^0_{DS}(C(\mathfrak{n}) \otimes V_k(\bar{\mathfrak{g}})),$$

called the universal $\mathcal{W}$-algebra associated to $\bar{g}$ of level k.
(Arakawa) All non-zero cohomology groups vanish.

$$\mathcal{W}_k(\bar{\mathfrak{g}}) := \mathcal{W}^k(\bar{\mathfrak{g}})/I_{max}$$

If $k = -h^\vee + \frac{p}{q}$, $p, q \in \mathbb{N}$, $p \geq h, q \geq h^\vee$, $(p, q) = 1$, then $c(k) = l(1 - h^\vee(h^\vee + 1)\frac{(p-q)^2}{pq})$.

Conjecture

Vertex algebra $\mathcal{W}_k(\bar{\mathfrak{g}})$ is strongly rational if and only if it is C_2-cofinite, which is if and only if k is as above.

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Vertex algebra $\mathcal{W}_k(\bar{\mathfrak{g}})$ is strongly rational if and only if it is C_2-cofinite, which is if and only if k is as above.

Quantum DS-reduction: $\mathfrak{g} = sl_2$

Example $k = -2 + \frac{p}{q}$, $(p, q) = 1$, $p, q \geq 2$. Then $\mathcal{W}_k(\bar{\mathfrak{g}})$ is the irreducible highest weight module for the Virasoro algebra $L(c_{p,q}, 0)$ with central charge $c_{p,q} = 1 - \frac{6(p-q)^2}{pq}$.

It is well-known that $L(c_{p,q}, 0)$ is C_2-cofinite and strongly rational. For other values of c, $L(c, 0)$ is not C_2-cofinite.

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Lattice vertex algebras

$(L, \langle\ \ \rangle)$ - a positive definite even integral lattice and $\mathbb{C}[L]$ its group algebra.

$$\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C},$$

$\hat{\mathfrak{h}}$ - the corresponding Heisenberg algebra.

$$\widehat{\mathfrak{h}} = \widehat{\mathfrak{h}}_{<0} \oplus \mathfrak{h} \oplus \mathbb{C}K \oplus \widehat{\mathfrak{h}}_{>0}.$$

$M(1) \cong S(\hat{\mathfrak{h}}_{<0})$- the Fock space for $\widehat{\mathfrak{h}}$.
The space

$$V_L = M(1) \otimes \mathbb{C}[\hat{L}],$$

where $\hat{L}$ is a suitable extension of L , has a natural vertex algebra structure such that

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Naive approach

Start from a set of primary vectors $\{v_1, ..., v_k\}$, and consider

$$\mathcal{W} := \langle v_1, ..., v_k \rangle \subset V_L.$$

Yet, if $v_i = e^{\beta_i}$ and $\{\beta_i\}$ is a $\mathbb{Z}$-basis of L there is a nice combinatorial description of $\mathcal{W}$ [MP].

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Interesting $\mathcal{W}$-algebras from lattice vertex algebras

The main case of interest: $L = \sqrt{p}Q$, $p \geq 2$, and Q is as before. Choose the Virasoro element ω such that

$$c = l(1 - h^\vee(h^\vee + 1)\frac{(p-1)^2}{p})$$

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The vertex algebra $\mathcal{W}(p)_Q$ is C_2-cofinite.

Conjecture (FT)

There is an equivalence of (at least abelian) categories:

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- The triplet vertex algebra $\mathcal{W}(p)$ was introduced in [FHST], and $W(2)$ earlier in [GK].
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The case of mixed screenings

Consider V_L, with $L = \mathbb{Z}\alpha$, $\langle \alpha, \alpha \rangle = 2pp'$. Choose

$$\omega_{p,p'} = \frac{\alpha(-1)^2}{4pp'} + \frac{p-p'}{2pp'}\alpha(-2),$$

such that the central charge is $1 - 6\frac{(p-p')^2}{pp'}$. We have two screenings:

$$Q = e_0^{\alpha/p'} \quad \text{and} \quad \tilde{Q} = e_0^{-\alpha/p}$$

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Compared to $\mathcal{W}(p)$, the vertex operator algebra $\boldsymbol{\mathcal{W}}_{p,p'}$ brings several new features [FGST3-4] ,[AM2010].
(i) $\boldsymbol{\mathcal{W}}_{p,p'}$ is not simple, and it contains a unique vertex ideal $\mathcal{R}_{p,p'}$ such that

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The structure of $\mathcal{W}_{2,p}$

Conjecture

(i) $\mathcal{W}_{p,p'}$ is C_2-cofinite and strongly generated by 3 primary vectors.
(ii) [FGST3-4] $\mathcal{W}_{p,p'}$ has precisely $2pp' + \frac{(p-1)(p'-1)}{2}$ irreps.
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Theorem (AM 2010,2011)

The conjecture holds for $p' = 2$ and all p.

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where

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Logarithmic modules

Although the triplet $\mathcal{W}(p)$ (and "cousins") came from a lattice vertex algebra, it is difficult to construct its indecomposable modules. In particular, there is no obvious construction of projective covers of irreducible modules. These modules are often logarithmic (i.e. the action of $L(0)$ is nondiagonalizable).

In [AM] and [NT], logarithmic modules are constructed by using screening operators by a *deformation* of the action of $W(p)$ on the direct sum of two V_L-modules. $\mathcal{W}(p)$ admits $L(0)$-Jordan blocks of size at most 2.

Theorem (AM)

There are logarithmic $\mathcal{W}_{p,p'}$-modules of $L(0)$-nilpotency rank 3, which is the highest nilpotency rank. These modules are realized inside direct sum of four irreducible V_L-modules.

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In [AM] and [NT], logarithmic modules are constructed by using screening operators by a *deformation* of the action of $W(p)$ on the direct sum of two V_L-modules. $\mathcal{W}(p)$ admits $L(0)$-Jordan blocks of size at most 2.

Theorem (AM)

There are logarithmic $\mathcal{W}_{p,p'}$-modules of $L(0)$-nilpotency rank 3, which is the highest nilpotency rank. These modules are realized inside direct sum of four irreducible V_L-modules.

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Example

"Powers" of screening operators (for $\mathcal{W}(p)$). Take the residue of

$$Q(z) = Y(e^{-\alpha/p}, z)$$

$$Q^{[2]}(z) = \sum_{m\geq 0} e^{-\alpha/p}_{-m-1/p} z^{m+1/p} \frac{p}{1+mp} \binom{-2/p}{m} Y(e^{-\alpha/p}, z)$$

$$+ \sum_{m\geq 0} \frac{p}{1+mp} \binom{-2/p}{m} Y(e^{-\alpha/p}, z) e^{-\alpha/p}_{m+1/p} z^{-m-1/p}.$$

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Higher rank case

Introduced in [FT].
$\mathcal{W}(p)_Q$ has a large "nilpotent" subalgebra $\mathcal{B}(p)_Q$.

The zero "charge" subalgebra:

$$M(1) \supset \mathcal{W}_p(\mathfrak{g}) \subset \mathcal{W}(p)_Q^{(0)} \subset \mathcal{W}(p)_Q$$

Essentially, if suitable conditions on $\mathcal{W}(p)_Q^{(0)}$ are satisfied, $\mathcal{W}(p)_Q$ will be C_2-cofinite.

Theorem

The algebra $\mathcal{W}(p)_Q$ admits logarithmic modules of $L(0)$-nilpotent rank at least rank$(\mathfrak{g})$. This fits with the conjectural description of irreducible $\mathcal{W}(p)_Q$-characters.

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Conclusion

Lattice $\mathcal{W}$-algebras and their (logarithmic) modules.

- "Honest" representation theory (blocks, etc.)
- Kazhdan-Lusztig correspondence [FGST3-4]
- New examples of finite tensor categories [HLZ]
- Boundary conditions [GRW2009].
- Logarithmic modular forms
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THANK YOU !!!

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